Detecting Malaria Parasites in Thin-Blood-Smear Cell Images

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Problem

Diagnosing malaria from a thin blood smear depends on a trained person examining very large numbers of cells. I build and evaluate a classifier that labels a single segmented red-blood-cell image as parasitized or uninfected. The aim is to apply the full machine learning workflow to a medical-imaging problem and to evaluate it honestly. The system is a research demonstration and is explicitly **not** a diagnostic tool.

Data and rights

Dataset: "Cell Images for Detecting Malaria" (Kaggle, `iarunava/cell-images-for-detecting-malaria`), from the NIH/NLM malaria collection: 27,558 unique segmented cell images in two classes. Citation: Rajaraman et al., *PeerJ* 6:e4568 (2018).

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Personal data: none known. Filenames carry pseudonymous patient codes and capture timestamps, which I keep out of the demo.

Known issues: the download contains every image twice (I remove the copy), and cells from the same patient are correlated, so all splits are made by patient.

Success statement

The project succeeds if:

1. The model is evaluated on held-out **patients** (no patient appears in more than one split), and the test set is used only for final reporting.
2. It beats a simple colour-feature baseline on that test set, with a patient-level bootstrap interval for the difference that excludes zero.

3. I report sensitivity and specificity, specificity separately for negative cells from healthy and from infected patients, calibration, and a failure analysis, not accuracy alone.
4. Security, privacy and misuse risks are addressed and documented in a model card.
5. A stranger can clone the repository and run the demo and tests from the README.

Approach

Data audit and EDA, then a patient-grouped baseline (gradient boosting on colour features), then a fine-tuned ResNet18, then a small hyperparameter search on validation data only, then robustness, calibration and error analysis, then hardening (input validation, safetensors weights, pinned dependencies), then the demo and defence.

Safety, privacy and risk

The output is a probability for research use. The demo states that it is not diagnostic and withholds a prediction for images that do not resemble the training cells.

Main risks: a single dataset with no external validation, possible label noise, stain differences between patients that a model could exploit, and vulnerability to deliberately altered images. I measure these rather than assume them away.

Out of scope

Whole-slide detection, other parasite species, clinical validation, and any deployment.

Reference

Rajaraman S, Antani SK, Poostchi M, Silamut K, Hossain MA, Maude RJ, Jaeger S, Thoma GR. Pre-trained convolutional neural networks as feature extractors toward improved malaria parasite detection in thin blood smear images. PeerJ. 2018 Apr 16;6:e4568. doi: 10.7717/peerj.4568. PMID: 29682411; PMCID: PMC5907772.